Kampala Traffic Predictor

Professional Project Manual & Technical Blueprint

Version: 1.0

Last Updated: 2024

Architecture: Three-Tier Web Application

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1. Executive Summary

The **Kampala Traffic Predictor System** is a web-based application designed to predict and manage traffic congestion in Kampala, Uganda. The system leverages real-time traffic data, historical patterns, and predictive algorithms to help users avoid congested routes and make informed travel decisions.

Key Objectives

- Predict traffic congestion at major hotspots in Kampala
- Provide real-time traffic updates and alerts
- Suggest alternative routes during peak hours
- Enable user-generated traffic reports
- Visualize traffic patterns through interactive maps and analytics

Target Users

- **General Users:** Citizens who need traffic information for daily commutes
- **System Administrators:** Manage hotspots, approve reports, and analyze traffic data

2. System Architecture Overview

Architecture Type: Three-Tier Architecture

```
┌─────────────────────────────────────────────────────────────┐
│                      PRESENTATION LAYER                      │
│  (HTML, CSS, JavaScript, Bootstrap, Tailwind, jQuery)        │
└─────────────────────────────────────────────────────────────┘
                              ↕
┌─────────────────────────────────────────────────────────────┐
│                      APPLICATION LAYER                       │
│         (PHP Server-side Processing & API Endpoints)         │
└─────────────────────────────────────────────────────────────┘
                              ↕
┌─────────────────────────────────────────────────────────────┐
│                        DATA LAYER                            │
│              (MySQL Database - MariaDB/MySQL)                │
└─────────────────────────────────────────────────────────────┘
```

System Flow Diagram

```
User Login → Authentication → Dashboard → Feature Access
                     ↓
              Session Management
                     ↓
         Role-Based Access Control
                     ↓
    ┌────────────────┴────────────────┐
    ↓                                 ↓
Admin Features              User Features
- Manage Hotspots          - View Predictions
- Approve Reports          - Submit Reports
- View Analytics           - View Traffic Map
- Manage Users             - Route Suggestions
- System Settings          - Profile Management
```

3. Technology Stack

Frontend Stack

HTML5

- Semantic markup for structure
- Form validation attributes
- Responsive meta tags

CSS Frameworks

- **Bootstrap 5.3.0:** Component library, grid system, responsive utilities
- **Tailwind CSS 2.2.19:** Utility-first styling, custom design system

JavaScript Libraries

- **jQuery 3.6.0:** DOM manipulation, AJAX requests, event handling
- **Chart.js:** Data visualization for analytics
- **Google Maps API:** Interactive traffic map visualization

Custom Assets

- `custom.css` : Application-specific styling
- `validation.js` : Client-side form validation
- `ajax_requests.js` : AJAX utility functions
- `predictions.js` : Prediction formatting and display
- `traffic_map.js` : Map initialization and marker management

Backend Stack

PHP (Server-side)

- Session management

- User authentication
- Database operations
- API endpoints
- Role-based access control

Database

- **MySQL/MariaDB:** Relational database management
- **Connection:** MySQLi extension with prepared statements

Server Configuration

- **Apache Web Server**
- **mod_rewrite:** URL routing
- **mod_headers:** Security headers
- **mod_deflate:** Response compression
- **mod_expires:** Cache control

4. Database Structure

Entity Relationship Diagram

```
┌──────────────┐          ┌──────────────┐          ┌──────────────────┐
│    users     │─────────>│   hotspots   │<─────────│  traffic_data    │
└──────────────┘          └──────────────┘          └──────────────────┘
      │                         │
      │                         │
      ↓                         ↓
┌──────────────────┐     ┌──────────────────────┐
│ traffic_reports  │     │ traffic_predictions  │
└──────────────────┘     └──────────────────────┘

┌──────────────────┐
│ system_settings  │
└──────────────────┘
```

Table Structures

1. users Table

Field	Type	Description
id	INT AUTO_INCREMENT PRIMARY KEY	Unique user identifier
fullname	VARCHAR(100)	User's full name
email	VARCHAR(120) UNIQUE	Login email (unique)
password	VARCHAR(255)	bcrypt hashed password
role	ENUM('admin','user')	User access level
created_at	TIMESTAMP	Registration date

2. hotspots Table

Field	Type	Description
id	INT AUTO_INCREMENT PRIMARY KEY	Unique hotspot identifier
location_name	VARCHAR(150)	Hotspot location name
latitude	DECIMAL(10,7)	Geographic latitude
longitude	DECIMAL(10,7)	Geographic longitude
peak_start	TIME	Peak hour start time
peak_end	TIME	Peak hour end time
created_by	INT	Admin user ID (FK)
created_at	TIMESTAMP	Creation date

3. traffic_data Table

Field	Type	Description
id	INT AUTO_INCREMENT PRIMARY KEY	Unique record identifier
hotspot_id	INT	Related hotspot (FK)
congestion_level	VARCHAR(50)	Light/Moderate/Heavy
vehicle_speed	DECIMAL(5,2)	Average speed (km/h)
traffic_density	INT	Vehicles per area
recorded_at	TIMESTAMP	Measurement timestamp

4. traffic_predictions Table

Field	Type	Description
id	INT AUTO_INCREMENT PRIMARY KEY	Unique prediction ID
hotspot_id	INT	Related hotspot (FK)
predicted_level	VARCHAR(50)	Light/Moderate/Heavy
prediction_time	DATETIME	Prediction target time
confidence_score	DECIMAL(5,2)	Prediction confidence %
created_at	TIMESTAMP	Prediction generated

5. traffic_reports Table

Field	Type	Description
id	INT AUTO_INCREMENT PRIMARY KEY	Unique report ID
user_id	INT	Submitter user (FK)
hotspot_id	INT	Related hotspot (FK)
traffic_level	VARCHAR(50)	Reported congestion
description	TEXT	Report details
status	ENUM('pending','approved','rejected')	Review status
created_at	TIMESTAMP	Submission date

6. system_settings Table

Field	Type	Description
id	INT AUTO_INCREMENT PRIMARY KEY	Settings record ID
sidebar_color	VARCHAR(50)	Sidebar background color
font_family	VARCHAR(50)	System font family
font_size	VARCHAR(20)	Base font size

5. Security Implementation

Authentication Security

Password Security

- **Hashing Algorithm:** bcrypt (PASSWORD_DEFAULT)
- **Salt:** Auto-generated per password
- **Cost Factor:** PHP default (currently 10)

```
// Password hashing on registration
$password = password_hash($_POST['password'], PASSWORD_DEFAULT);

// Password verification on login
password_verify($password, $user->password)
```

Session Management

- PHP sessions with secure configuration
- Session variables: user_id, fullname, email, role
- Session validation on every protected page
- Automatic session destruction on logout

Authorization Security

Role-Based Access Control (RBAC)

Feature	Admin	User
View Dashboard	✓	✓
Manage Hotspots	✓	✗
Approve Reports	✓	✗
Submit Reports	✓	✓
View Analytics	✓	✗
Manage Users	✓	✗
System Settings	✓	✗
View Predictions	✓	✓
View Traffic Map	✓	✓
Route Suggestions	✓	✓

Input Validation & Sanitization

Server-Side Validation

```
// Email sanitization
filter_var($_POST['email'], FILTER_SANITIZE_EMAIL)

// String sanitization
filter_var($_POST['name'], FILTER_SANITIZE_STRING)

// SQL injection prevention (Prepared Statements)
$stmt = mysqli_prepare($conn, $query);
mysqli_stmt_bind_param($stmt, "ss", $email, $password);
```

Client-Side Validation

```
// Email validation regex
function validateEmail(email) {
    const re = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    return re.test(email);
}

// Password strength validation
function validatePassword(password) {
    return password.length >= 6;
}
```

HTTP Security Headers

- **X-Frame-Options:** SAMEORIGIN (prevent clickjacking)
- **X-Content-Type-Options:** nosniff (prevent MIME sniffing)
- **X-XSS-Protection:** 1; mode=block (XSS protection)
- **Referrer-Policy:** strict-origin-when-cross-origin
- **Content-Security-Policy:** Restrictive CSP rules

Directory & File Protection

- Directory listing disabled (Options -Indexes)
- Sensitive directories blocked: includes/, config/, database/
- Hidden files blocked (files starting with .)
- HTTPS enforcement with automatic redirect

6. Real-Time Features

Auto-Refresh Mechanisms

Dashboard Statistics (30-second interval)

```
$(document).ready(function() {
    loadDashboardStats();
    setInterval(loadDashboardStats, 30000); // 30 seconds
});
```

Traffic Predictions (60-second interval)

```
$(document).ready(function() {
    loadPredictions();
    setInterval(loadPredictions, 60000); // 1 minute
});
```

AJAX-Based Real-Time Updates

- Asynchronous requests using jQuery AJAX
- Background data updates without page refresh
- Dynamic DOM manipulation for seamless UX

Traffic Prediction Algorithm

```
Input: Historical traffic data, current time, peak hours
Process:
  1. Fetch average speed and density from last hour
  2. Apply congestion thresholds:
     - Heavy: speed < 10 km/h
     - Moderate: speed < 25 km/h
     - Light: speed ≥ 25 km/h
  3. Check if current time is within peak hours
  4. Calculate confidence score (75-90%)
Output: Predicted congestion level with confidence
```

Real-Time Map Updates

- Dynamic marker placement based on hotspot data
- Interactive info windows with real-time information
- Traffic layer overlay for live traffic conditions
- Auto-refresh of marker data

Instant Report Submission

- User-submitted reports immediately stored
- Admin dashboard updated in real-time
- Status changes reflect instantly across all admin sessions

7. Frontend Architecture

Design System

Color Palette

- **Primary:** Bootstrap Blue (#0d6efd)

- **Success:** Green (#198754)
- **Warning:** Yellow (#ffc107)
- **Danger:** Red (#dc3545)
- **Info:** Cyan (#0dcaf0)
- **Dark Sidebar:** Customizable (default #343a40)

Typography

- **Font Family:** Customizable (Arial, Helvetica, Times New Roman, Georgia)
- **Font Size:** Customizable (12px, 14px, 16px, 18px)
- **Line Height:** Standard Bootstrap scale

Responsive Breakpoints

- **Mobile:** < 768px (Sidebar becomes full-width)
- **Tablet:** 768px+ (Standard sidebar)
- **Desktop:** 992px+ (Optimized layout)
- **Large:** 1200px+ (Maximum width utilization)

Component Structure

Layout Components

- **Sidebar Navigation:** Fixed left sidebar with role-based menu items
- **Content Area:** Flexible main content with margin-left offset
- **Header:** Dynamic with system settings integration
- **Footer:** Script imports and initialization

Data Display Components

- **Statistics Cards:** Dashboard metrics with color coding
- **Data Tables:** Sortable, responsive tables with action buttons
- **Charts:** Chart.js visualizations (bar, line, heatmap)
- **Maps:** Google Maps with custom markers and info windows
- **Alerts:** Bootstrap alerts for predictions and notifications

Form Components

- **Login/Register Forms:** Validation with error messaging
- **Modal Forms:** Bootstrap modals for add/edit operations
- **Inline Forms:** Route suggestions, profile updates

8. Backend Architecture

PHP File Organization

Page Controllers (Root Level)

- login.php, register_user.php, register_admin.php
- dashboard.php, profile.php
- hotspots.php, users.php, settings.php
- traffic_map.php, predictions.php, routes.php
- traffic_reports.php, analytics.php

API Endpoints (api/ directory)

File	Purpose
login.php	User authentication
register_user.php	User registration
register_admin.php	Admin registration
fetch_hotspots.php	Get all hotspots
add_hotspot.php	Create new hotspot
delete_hotspot.php	Remove hotspot
fetch_reports.php	Get traffic reports
submit_report.php	Create new report
approve_report.php	Approve pending report
traffic_prediction.php	Generate predictions
fetch_dashboard_stats.php	Dashboard metrics
update_profile.php	User profile update
update_settings.php	System settings update

API Response Format

Success Response

```
{
    "success": true,
    "message": "Operation completed successfully",
    "data": { ... }
}
```

Error Response

```
{
    "success": false,
    "message": "Error description",
    "data": null
}
```

Database Access Pattern

1. Prepare statement: mysqli_prepare($conn, $query)
2. Bind parameters: mysqli_stmt_bind_param($stmt, "types", $vars)
3. Execute: mysqli_stmt_execute($stmt)
4. Get result: mysqli_stmt_get_result($stmt)
5. Fetch data: mysqli_fetch_object($result)
6. Return JSON: json_encode(['success' => true, 'data' => $data])

9. API Documentation

Authentication APIs

POST /api/login.php

Description: Authenticate user credentials

Request Body:

```
{
    "email": "user@example.com",
    "password": "userpassword"
}
```

Response:

```
{
    "success": true,
    "message": "Login successful"
}
```

POST /api/register_user.php

Description: Register new user account

Request Body:

```
{
    "fullname": "John Doe",
    "email": "john@example.com",
    "password": "password123"
}
```

Hotspot Management APIs

GET /api/fetch_hotspots.php

Description: Retrieve all traffic hotspots

Response:

```
{
    "success": true,
    "data": [
        {
            "id": 1,
            "location_name": "Clock Tower",
            "latitude": "0.3136",
            "longitude": "32.5811",
            "peak_start": "07:00:00",
            "peak_end": "09:00:00"
        }
    ]
}
```

POST /api/add_hotspot.php

Authorization: Admin only

Request Body:

```
{
    "location_name": "Ntinda",
    "latitude": "0.3555",
    "longitude": "32.6178",
    "peak_start": "17:00",
    "peak_end": "19:00"
}
```

Traffic Report APIs

POST /api/submit_report.php

Authorization: Authenticated users

Request Body:

```
{
    "hotspot_id": 3,
    "traffic_level": "Heavy",
    "description": "Accident at intersection"
}
```

Prediction APIs

GET /api/traffic_prediction.php

Description: Generate and retrieve traffic predictions

Response:

```
{
    "success": true,
    "data": [
        {
            "location_name": "Clock Tower",
            "predicted_level": "Heavy",
            "prediction_time": "2024-01-15 09:45:00",
            "confidence_score": 85.5
        }
    ]
}
```

10. User Roles & Access Control

Admin Role Capabilities

- Dashboard Access: View all system statistics
- Hotspot Management: CRUD operations on traffic hotspots
- Report Management: Approve/reject user-submitted reports
- User Management: Create, update, delete user accounts
- Analytics Access: View traffic trends and charts
- System Settings: Customize UI appearance
- All User Features: Can perform all user-level actions

User Role Capabilities

- Dashboard Access: View personal statistics
- Report Submission: Submit traffic reports for approval
- Traffic Map: View real-time traffic visualization
- Predictions: View AI-generated traffic forecasts
- Route Suggestions: Get alternative route recommendations
- Profile Management: Update personal information

Access Control Implementation

```
// Page-level protection
<?php
require_once 'includes/admin_auth.php'; // Admin pages
require_once 'includes/auth.php';       // All authenticated pages
?>

// API-level protection
if ($_SESSION['role'] !== 'admin') {
    echo json_encode(['success' => false, 'message' => 'Unauthorized']);
    exit();
}
```

11. System Features

1. User Authentication System

- Registration: Separate flows for users and admins
- Login: Email and password authentication
- Session Management: Persistent login across pages
- Logout: Secure session destruction

2. Traffic Hotspot Management

- Create Hotspots: Define location name, coordinates, peak hours
- View Hotspots: Table view with all details
- Update Hotspots: Modify existing hotspot information
- Delete Hotspots: Remove outdated hotspots
- Map Visualization: See all hotspots on Google Maps

3. Traffic Prediction Engine

- Algorithm: Statistical analysis of historical data
- Peak Hour Detection: Automatic heavy traffic during peak windows
- Confidence Scoring: 75-90% confidence levels
- Real-Time Updates: Predictions refresh every 60 seconds
- Visual Indicators: Color-coded alerts (red=heavy, yellow=moderate, green=light)

4. User-Generated Reports

- Report Submission: Users describe current traffic conditions
- Approval Workflow: Admin review before publication
- Status Tracking: Pending, approved, rejected states
- Integration: Approved reports influence predictions

5. Interactive Traffic Map

- Google Maps Integration: Full-screen interactive map
- Hotspot Markers: Clickable pins at each location
- Info Windows: Display peak hours and details
- Traffic Layer: Real-time Google traffic overlay

6. Route Suggestions

- Input: Select congested hotspot
- Output: Alternative route recommendations
- Time Savings: Estimated time saved by avoiding congestion
- Multiple Options: 2-3 alternative routes per query

7. Analytics Dashboard

- Chart Types: Bar charts, line graphs, trend analysis

- Metrics: Most congested roads, peak hours, trends
- Time Ranges: Daily, weekly, monthly views

8. System Customization

- Sidebar Color: Admin-configurable color scheme
- Font Family: Choose from 4 font options
- Font Size: Adjustable text size (12-18px)
- Live Preview: Settings apply immediately after save

12. Installation & Setup

Prerequisites

- **Web Server:** Apache 2.4+
- **PHP:** 7.4 or higher
- **Database:** MySQL 5.7+ or MariaDB 10.3+
- **Extensions:** mysqli, session, json, filter

Step 1: Database Setup

```
# Create database and import schema
mysql -u root -p

SOURCE database/schema.sql;
SOURCE database/sample_data.sql;  -- Optional
```

Step 2: Configure Database Connection

Edit `config/database.php`:

```
$host = 'localhost';
$username = 'your_db_username';
$password = 'your_db_password';
$database = 'kampala_traffic_predictor';
```

Step 3: Configure Google Maps API

Edit `traffic_map.php`:

```
<script src="https://maps.googleapis.com/maps/api/js?key=YOUR_API_KEY"><
```

Step 4: Enable Apache Modules

```
sudo a2enmod rewrite
sudo a2enmod headers
sudo a2enmod deflate
sudo a2enmod expires
sudo systemctl restart apache2
```

Step 5: Access the Application

Navigate to: `http://localhost/kampala-traffic/`

Default Login Credentials (Sample Data)

- **Admin:** admin@traffic.com / password
- **User:** john@example.com / password

13. File Structure

```
kampala-traffic-predictor/
│
├── index.php                    # Redirects to login
├── login.php                    # User login page
├── logout.php                   # Session destruction
├── register_user.php            # User registration
├── register_admin.php           # Admin registration
├── dashboard.php                # Main dashboard
├── profile.php                  # User profile
├── hotspots.php                 # Hotspot management
├── users.php                    # User management
├── settings.php                 # System settings
├── traffic_map.php              # Interactive map
├── predictions.php              # Traffic predictions
├── routes.php                   # Route suggestions
├── traffic_reports.php          # Report management
├── analytics.php                # Traffic analytics
├── .htaccess                    # Apache config
│
├── api/                         # API endpoints
│   ├── login.php
│   ├── register_user.php
│   ├── fetch_hotspots.php
│   ├── add_hotspot.php
│   ├── submit_report.php
│   ├── traffic_prediction.php
│   └── ...
│
├── includes/                    # Reusable components
│   ├── auth.php
│   ├── admin_auth.php
│   ├── header.php
│   ├── footer.php
│   └── sidebar.php
│
├── config/                      # Configuration
│   └── database.php
```

```
│
├── database/                    # Database schema
│   ├── schema.sql
│   └── sample_data.sql
│
└── assets/                      # Frontend assets
    ├── css/
    │   └── custom.css
    └── js/
        ├── validation.js
        ├── ajax_requests.js
        ├── predictions.js
        └── traffic_map.js
```

14. Security Best Practices

Current Implementation

- ✅ Password Hashing: bcrypt algorithm
- ✅ SQL Injection Prevention: 100% prepared statements
- ✅ XSS Protection: htmlspecialchars() on all output
- ✅ CSRF Protection: Session-based verification
- ✅ HTTPS Enforcement: Automatic redirect
- ✅ Security Headers: X-Frame-Options, CSP, etc.
- ✅ Directory Protection: Sensitive folders blocked
- ✅ Session Security: Secure session management
- ✅ Input Validation: Server and client-side
- ✅ Error Suppression: No database errors exposed

Recommended Enhancements

- ☐ CSRF Tokens: Implement token-based CSRF protection
- ☐ Rate Limiting: Prevent brute force attacks
- ☐ 2FA: Two-factor authentication for admins
- ☐ Password Policy: Enforce strong passwords
- ☐ Account Lockout: Lock after failed login attempts
- ☐ Audit Logging: Log all admin actions
- ☐ API Rate Limiting: Throttle API requests
- ☐ Database Encryption: Encrypt sensitive fields

15. Future Enhancements

Phase 1: Core Improvements

- Mobile Application: Native Android/iOS apps
- Real Traffic Data Integration: Government traffic API
- Machine Learning: Advanced prediction algorithms
- Notifications: Push notifications for alerts
- Weather Integration: Factor weather into predictions

Phase 2: Advanced Features

- User Preferences: Saved routes and favorites
- Social Features: Share routes, comment on reports
- Gamification: Points for accurate reports
- Multi-Language: Support for local languages
- Offline Mode: Cache predictions for offline access

Phase 3: Enterprise Features

- API for Third Parties: Public traffic data API
- Fleet Management: Integration with transport companies
- Government Dashboard: City planning insights
- Historical Analysis: Long-term trend reports
- Incident Management: Integration with emergency services

End of Project Manual

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